American Computer Science League

2025-2026 ● Contest 4: Short Problems Solutions ● Junior Division

1. Graph Theory

Squaring the adjacency matrix for paths of length 2 yields the following:

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

There are 13 entries of 1.

E. 13

2. Graph Theory

By inspection, there are 2 cycles in this directed graph. They are: ABA, CDC.

C. 2

3. Digital Electronics

The equivalent Boolean algebra expression is: $AB + (\overline{B + C})C$
It simplifies as follows:

$AB + (\overline{B + C})C$
$= AB + (\overline{B} * \overline{C})C$
$= AB + B\overline{C}C$
$= AB + 0$
$= AB$

D. AB

4. Digital Electronics

The equivalent Boolean algebra expression is: $\overline{AB} + (B + C)C$.
It simplifies as follows:
$\overline{AB} + (B + C)C$
$= \overline{A} + \overline{B} + BC + C$
$= \overline{A} + \overline{B} + C$
This is FALSE for (1,1,0) only. Therefore, 7 make it TRUE.

D. 7

5. What Does This Program Do? (Strings)

```
s = "742"
r = ""
for x = 0 to 2
    r = s[x] + r
next x
a = str(val(s) - val(r))
b = ""
for x = len(a)-1 to 0 step -1
    b = b + a[x]
next x
output val(b) + val(a)
```

This program reverses a 3-digit number, subtracts it from the original number, reverses the answer and adds these two together to always get 1089. In this program s="742" and r ="247". The value of a = 742 - 247 = 495. Then b=594 and the program outputs 495+594= 1089.

D. 1089

6. What Does This Program Do? (Strings)

```
s = "APCompSciAinJava"
ans = ""
for x = 0 to len(s) - 2 step 2
  if s[x] < s[x+1] then
    ans = ans + s[x] + s[x+1]
  else
    ans = ans + s[x+1] + s[x]
  end if
next x
output ans
```

This program looks at pairs of characters and puts them in alphabetical order by concatenating them into a new string.

B. APCompScAiinJaav